Clinical Decision Making and Pattern Recognition in Health Care

Agentic Generative AI for Treatment, Payment, and Operations

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Defining the Topic

In healthcare, clinical decision-making fundamentally relies on recognizing complex patterns across transactional claims, clinical documentation, and longitudinal patient encounters. Practically speaking, enterprise decision-makers must continuously distinguish normal clinical variance from improper or anomalous activity that requires human intervention. While traditionally associated with bedside care, this dynamic drives the Treatment, Payment, and Operations (TPO) triad—specifically within Payment Integrity and Utilization Review. Given the massive scale and complexity of modern healthcare data, organizations increasingly rely on machine learning to enable faster, more consistent adjudication. The most pivotal trend in this domain is the transition from passive, single-shot predictive models to Agentic Generative AI: architectures comprising autonomous Large Language Model (LLM) agents that actively investigate records rather than merely scoring them.

For Cotiviti, the highest-yield application of this paradigm is a Human-in-the-Loop (HITL) Adjudication Layer. Rather than issuing autonomous payment verdicts, this system flags aberrant utilization behavior, pre-pay DRG upcoding, or operational spikes before they consume analyst time. The model ranks records by financial risk and generates an auditable, narrative summary explaining why each item was surfaced. This design aligns perfectly with healthcare's strict mandate for transparency, reviewability, and trust—grounding AI efficiency in expert human judgment.

Relevant Trends

Rapid adoption of agentic AI across enterprise healthcare: Industry surveys indicate that over 80% of health care executives expect agentic and generative AI to deliver moderate-to-significant value across clinical, business, and back-office functions in 2026, with nearly all surveyed leaders anticipating measurable cost savings within the next reporting cycle. Health plans are aggressively transitioning from theoretical experimentation to live deployments in utilization management and prior authorization. Furthermore, provider-side comfort regarding AI-assisted authorization decisions is reportedly high, provided robust, transparent safeguards are maintained.

Maturing Fraud, Waste, and Abuse (FWA) analytics: Payment integrity vendors are combining supervised classification, unsupervised anomaly detection, and reinforcement learning to score claims dynamically at submission and re-score them as new clinical data arrives later in the claim lifecycle. Established enterprise platforms now process entire payer claim populations in batch or real-time rather than relying on random sampling. Crucially, these systems pair statistical anomaly scores with explicit reason codes so human investigators can act rapidly on flagged cases.

A calibrated, risk-tiered regulatory environment: The FDA's updated 2026 Clinical Decision Support guidance grants software greater latitude to offer single clinically appropriate recommendations without automatically triggering strict medical-device regulation, provided the tool remains transparent and reviewable by a clinician. However, the agency has left the statutory term "clinically appropriate" undefined and continues to strictly regulate any function interpreting medical imaging, signaling that federal oversight will remain calibrated to patient risk rather than uniformly relaxed.

Opportunities and Threats

Opportunities

- **Enhanced FWA Detection Precision:** An ensemble approach pairing statistical anomaly detection with generative reasoning discovers novel, zero-day upcoding schemes that evade rigid rules-based logic—directly strengthening Cotiviti's primary payment integrity offering.
- **Autonomous Audit Triaging:** Specialized agents capable of autonomously retrieving prerequisite chart history, past claims, and coverage policies can pre-package an audit file for human analysts, drastically shrinking review times without removing human oversight.
- **Verdict Explainability as a Competitive Moat:** Because payer clients increasingly demand clear logic behind automated payment determinations, building native explainability into the software roadmap serves as a major market differentiator.

Threats

- **Regulatory Ambiguity:** Undefined statutory phrases such as “clinically appropriate” escalate compliance exposure for any system whose recommendations could be interpreted by regulators as autonomous decision-making.
- **Adversarial Model Gaming:** As automated FWA detection thresholds become understood by bad actors, perpetrators will abuse their own generative AI tools to systematically craft synthetic claims optimized to bypass Cotiviti's detection filters.
- **Automation Bias among Human Reviewers:** As AI-generated investigative summaries appear increasingly authoritative, human reviewers risk falling into passive complacency—relying entirely on the AI’s conclusions even when underlying source evidence is sparse.

Strategic Options for Cotiviti

- **Deploy the Cotiviti Agentic Audit Copilot:** Invest in an autonomous claims-investigation assistant that pre-assembles the supporting evidence (longitudinal provider patterns, CPT-to-ICD mapping, policy language) behind every anomaly flag. This drastically reduces investigator prep time while reserving final payment sign-off strictly for certified human analysts.
- **Architect Native, Tiered Explainability:** Build audit-ready reason codes into the core product roadmap today rather than treating explainability as a reactionary compliance fix—putting Cotiviti ahead of emerging FDA and payer transparency demands.
- **Pilot Online (Continuous) Anomaly Retraining:** Implement dynamic, continuous learning models on high-risk claim categories most susceptible to manipulation, justifying a strategic transition from static rules-based anomaly detection to adaptive AI.

Conclusion

The development of agentic generative AI moves healthcare administrative pattern recognition away from passive, one-time recommendations toward active, continuous investigation. For a payment integrity firm, this evolution brings immense leverage alongside distinct operational risks. While autonomous orchestration offers an unprecedented ability to optimize investigator workflows and curb emerging fraud schemes, it simultaneously demands bulletproof decision explainability and heightened defenses against adversarial AI manipulation.

Bibliography

Deloitte. “Many Health Care Leaders Are Leaning into Agentic AI as Adoption Hurdles Ease.” Deloitte Insights, 16 Apr. 2026, www.deloitte.com/us/en/insights/industry/health-care/agentic-ai-health-care-operating-model-change.html.

Faegre Drinker Biddle & Reath LLP. “Key Updates in FDA’s 2026 General Wellness and Clinical Decision Support Software Guidance.” Faegre Drinker, Jan. 2026, www.faegredrinker.com/en/insights/publications/2026/1/key-updates-in-fdas-2026-general-wellness-and-clinical-decision-support-software-guidance.

Gorenshtein, A., et al. “Agentic AI, Medical Morality, and the Transformation of the Patient-Physician Relationship.” arXiv, 2026, arxiv.org/pdf/2602.16553.

H2O.ai. “Claims Fraud Detection.” H2O.ai, h2o.ai/solutions/use-case/claims-fraud-detection/. Accessed 28 June 2026.

KevinMD. “FDA Loosens AI Oversight: What Clinicians Need to Know About the 2026 Guidance.” KevinMD, 18 Jan. 2026, kevinmd.com/2026/01/fda-loosens-ai-oversight-what-clinicians-need-to-know-about-the-2026-guidance.html.

Lone Star Analytics et al. “AI- and Data Science–Driven Healthcare Revenue and Payment Integrity Optimization.” Frontiers in Computer Science and AI, 9 Mar. 2026, al-kindipublishers.org/index.php/fcsai/article/download/12249/10945/33112.

Luo, Wei, and Marcus Gallagher. “Unsupervised DRG Upcoding Detection in Healthcare Databases.” 2010 IEEE International Conference on Data Mining Workshops, IEEE, 2010, pp. 600–605.

Philips. “Emerging Healthcare AI Trends in 2026 So Far.” Philips, 30 Apr. 2026, www.philips.com/a-w/about/news/archive/features/2026/this-years-emerging-trends-in-healthcare-ai-so-far.html.

SAS Institute. “SAS Payment Integrity for Health Care.” SAS, www.sas.com/en_us/software/payment-integrity-for-health-care.html. Accessed 28 June 2026.